10. $11 \times (-1) =$ _____

(i) –1 (ii) 0 (iii) +1 (iv) –11

11. $(-12) \times (-9) =$ _____

(i) 108 (ii) –108 (iii) +1 (iv) –1

1.5 Division of Integers

We have already seen that each multiplication statement has two division statements in whole numbers. Can we try the same for integers also; let us play with this number converter machine and see how division works.

Number Converter

Input	Multiplication Statement	Division Statement	Output
(i) Product of positive integer and positive integer	$8 \times 9 = 72$	$72 \div 9 = 8$, $72 \div 8 = 9$	Positive integer divided by Positive integer is Positive
(ii) Product of positive integer and negative integer	$-10 \times 7 = -70$	$(-70) \div 7 = -10$, $(-70) \div (-10) = 7$	Negative integer divided by Positive integer is Negative Negative integer divided by Negative integer is Positive
(iii) Product of negative integer and negative integer	$(-5) \times (-9) = 45$	$45 \div (-9) = -5$, $45 \div (-5) = -9$	Positive integer divided by Negative integer is Negative

From the above table, we infer that the division of two integers with the same sign is a positive integer.

Division of two integers with opposite signs gives a negative integer.

Try these

(i) $(-32) \div 4 =$ _____ (ii) $(-50) \div 50 =$ _____ (iii) $30 \div 15 =$ _____

(iv) $-200 \div 10 =$ ____ (v) $-48 \div 6 =$ ____

Example 1.25 Divide: (i) (–85) by 5 (ii) (–250) by (–25) (iii) 120 by (–6) (iv) 182 by (–2)

Solution (i) $(-85) \div 5 = -17$ (ii) $(-250) \div (-25) = +10$

(iii) $120 \div (-6) = -20$ (iv) $182 \div (-2) = -91$

Example 1.26

In a competitive exam 4 marks are given for every correct answer and (-2) marks are given for every incorrect answer, kalaivizhi attended the exam and answered all the questions and scored 20 marks only even though she got 10 correct answers. How many questions did she answer incorrectly?



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